

Wawayu: Offering glitchless gaming and app experiences with abundant network resources around the world

By leveraging the
content delivery r
and rich GPU res
Google Cloud, W
able to offer stab

Tell us your challenge. We're here to help.

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ABOUT WAWAYU



About Wawayu Network Technology

Founded in 2016 in Shenzhen, Wawayu Network Technology is a fast-growing internet company focusing on overseas markets with offices in Hong Kong and Guangzhou. Specialized in monetizing entertainment content on mobile devices, Wawayu has developed and distributed hundreds of apps ranging from gaming, keyboard to social networking while offering app marketing solutions. It currently has more than 500 million users across Europe, North America, South America, and Southeast Asia.

glitchless gaming experiences around the world and build a generated content application in a short amount of time.

Industries: Games

Location: China

Balancing

Google Cloud (<https://cloud.google.com/>),
[Cloud APIs](https://cloud.google.com/apis/) (<https://cloud.google.com/apis/>),

Cloud CDN (<https://cloud.google.com/cdn/>)

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

Cloud APIs (<https://cloud.google.com/apis/>)

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Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Google Cloud results

- Simplifies gaming content distribution setup for higher work efficiency with Google Cloud APIs
- Enhances content delivery speed for lower latency through Cloud CDN

Developed a generated content application within two months

- **Supports
glitchless
gaming
experience
during high
traffic
times with
Cloud Load
Balancing**

As the competition in China's mobile gaming market becomes intense, more and more Chinese gaming app developers have increased their investments in overseas markets. Facing different business environments from China, they need additional resources and technical support to successfully promote their products abroad.

Wawayu Network Technology is specialized in app monetization and marketing solutions for markets outside China. Founded in 2016 in Shenzhen, Wawayu first released iMoreKeyboard, a keyboard application supporting 144 languages that once had more than 10 million users worldwide. Following the success, the company expanded its business range and launched several apps ranging from gaming to social networking while gathering experiences in promoting application products in overseas markets. Now, Wawayu's business focus lies in distributing gaming apps outside China using its own app marketing solutions designed for international platforms to help other Chinese app developers gain global users more effectively.

After the launch of a gaming app, a developer normally would advertise the product across various social media platforms and websites to increase the app's download and purchase numbers. To enhance the product promotion result, Wawayu offers a comprehensive set of solutions, including app release, paid download, in-app purchase, advertising, subscription, and payment. For example, one of its solutions monitors the download numbers and growth trends of games of different categories to help app developers identify the best ways to increase their revenues.

Using its app promotion solutions, Wawayu has realized exponential growth in revenues with its self-developed apps. Together with applications that it helps distribute, Wawayu currently operates 500+ apps that have more than 500 million users across Europe, North America, South America, and Southeast Asia.

"The advantage of our app marketing solutions is that we use our self-built business intelligence system to conduct data analytics and enhance conversion rates," notes Yaobin Liu, Co-Founder and Chief Software Architect at Wawayu. "Through this, we have not only increased the revenues of our own apps but also helped other game developers promote their products and gain users more effectively."

At first, Wawayu used two content delivery networks to offer app services worldwide. However, the company later found that two networks were still not enough to constantly offer good connectivity,

especially for users in the regions with less ideal internet infrastructures like South Asia and Southeast Asia. To ensure that its users across the world can smoothly connect to its game servers at any time, Wawayu decided to incorporate Google Cloud (<https://cloud.google.com/>) into its IT infrastructure in 2022.

"The network quality and resource availability offered by different cloud platforms vary in different regions. To better ensure that all our users around the world can enjoy high-quality gaming experiences, we decided to add Google Cloud to our cloud infrastructure," says Liu. "Through a multi-cloud architecture, we can also prevent external risk factors like policy changes from disrupting our services."

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Enhancing global network quality with Cloud CDN and Cloud Load Balancing

Leveraging Google Cloud APIs

(<https://cloud.google.com/apis>), Wawayu swiftly
deployed the apps that it distributes outside China
on Google Cloud within two weeks. With Google
Cloud APIs like Cloud CDN

(<https://cloud.google.com/cdn>) and Cloud Load
Balancing (<https://cloud.google.com/load-balancing>),
Wawayu's staff are able to set up gaming content
distribution networks by themselves without the
help of its operations engineers, which has greatly
enhanced the team's work efficiency.

Now, Wawayu offers a network structure supported by three cloud platforms, including Google Cloud, to users around the world. Based on the current network status, users' devices would be connected to Wawayu's servers through the best-quality cloud network. Before incorporating Google Cloud into its cloud infrastructure, Wawayu sometimes was not able to ensure glitchless gaming experiences. By using Cloud CDN to deliver content from the nodes that are closest to end users and the widely distributed network nodes of Google Cloud, Wawayu has successfully improved the connectivity of its gaming apps in regions like Southeast Asia.

On top of that, Wawayu employs Cloud Load Balancing to distribute traffic across servers more evenly during high traffic times, which has further improved the user experiences that it offers. Liu points out that Google Cloud supports higher connection speed and stability. According to Wawayu's network data analytics report, the network of Google Cloud covers more than 39% of daily traffic among Wawayu's three networks and offers 12-17% lower latency compared to two other networks.

"The abundant network resources offered by Cloud CDN worldwide has effectively helped us enhance the connection speed of our games. Our end users' devices would automatically choose the best network to connect based on the location. Compared to other cloud platforms that we use, the usage of the Google Cloud network is the highest, which proves that it provides the best network quality," he adds.

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Accelerates development of AI-generated content applications with GKE

As the popularity of the AI-generated content (AIGC) technology increases, Wawayu has been exploring ways to combine the new technology with gaming and started developing a generative AI application that can transform short videos uploaded by users into cartoons. Since the application involves AI-related features, such as image-to-video and video display conversions, its development requires excellent computing power. Wawayu therefore decided to use Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) to support its development work. According to Liu, GKE provides richer GPU resources and wider model selection across regions compared to other offerings, especially the A100 GPU model that Wawayu relies on to develop this new product. As a result, Wawayu quickly built the app within two months.

"The generative AI technology is a popular field at the moment, and time to market is essential for us to snatch a leading position in the market," explains Liu. "Being able to use the rich computing resources of GKE to develop a new product in a short period of time has given us great confidence. We're now set to develop more generative AI applications that can meet the market's long-term needs and drive up our company's profits."

Expanding generative AI application and enhancing data analytics efficiency on Google Cloud

Besides developing more generative AI related products using Google Cloud, Wawayu also plans to move its big data workloads to BigQuery

(<https://cloud.google.com/bigquery>) in the future. The company currently uses different databases to support data storage and analytics, which results in repetitive work to synchronize databases. By merging its data storage and analytics workloads in BigQuery, Wawayu will be able to realize more efficient big data management.

"We want to leverage best-of-breed technologies to provide high-quality and innovative app products," says Liu. "Google Cloud has helped us significantly improve our service connectivity and development. Moving forward, we will leverage more Google Cloud products to continue realizing product optimization and innovation."

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